Exercise 1:	Data Visualization from CSV Using Python

Aim

To create a Python script that reads data from a CSV file containing social media post metrics and generates basic visualizations (bar chart, line chart, and pie chart) to analyze engagement across platforms and post types effectively.

Software Used

- Programming Language: Python 3.x
- Libraries:
  - Pandas (for data loading and manipulation)
  - Matplotlib (for creating visualizations)
- Environment: Jupyter Notebook.

Description

This exercise demonstrates how to visualize data from a CSV file using Python. The dataset contains information about social media posts, including platforms, post types, and engagement metrics (likes, shares, comments, views). The script:

- Loads and cleans the data (handles missing values and invalid entries).
- Aggregates metrics (e.g., average likes and views by platform).
- Generates three visualizations:
  - Bar chart: Compares average likes across platforms.
  - Line chart: Shows trends in average views across platforms.
  - Pie chart: Displays the percentage distribution of post types. These visualizations help analyze engagement patterns and distribution in the dataset.

Dataset Details

- **File Name**: social_media_data.csv
- **Location**: Place the file in the same directory as the script/notebook.
- **Columns**:
  - date: Date of the post (format: YYYY-MM-DD).

- platform: Social media platform (e.g., Twitter, Instagram, Facebook; includes some invalid entries like "NonExistingPlatform").
- post_type: Type of post (text, image, video).
- likes: Number of likes (may contain NaN/null).
- shares: Number of shares (may contain NaN/null).
- comments: Number of comments (may contain NaN/null).
- views: Number of views (may contain NaN/null).

- **Rows**: Approximately 500+ entries.

List of Tasks

1. Import necessary libraries (Pandas and Matplotlib).

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns
%matplotlib inline
```

2. Load the CSV file into a DataFrame.

```
df = pd.read_csv("social_media_data.csv")

df.head()
```

	date	platform	post_type	likes	shares	comments	views
0	2023-05-24	Twitter	image	868.0	238.0	352.0	8836.0
1	2023-02-23	Twitter	image	944.0	610.0	146.0	5554.0
2	2023-03-18	Instagram	image	1067.0	858.0	365.0	4018.0
3	2023-06-12	Twitter	text	780.0	427.0	100.0	6999.0
4	2023-02-03	Instagram	text	836.0	893.0	443.0	5802.0

3. Preview the data to verify loading.

```
print("dataset")
print(df)
```

```
Dataset:(7097)
            date   platform post_type   likes  shares  comments   views
0     2023-05-24    Twitter     image   868.0   238.0     352.0  8836.0
1     2023-02-23    Twitter     image   944.0   610.0     146.0  5554.0
2     2023-03-18  Instagram     image  1067.0   858.0     365.0  4018.0
3     2023-06-12    Twitter      text   780.0   427.0     100.0  6999.0
4     2023-02-03  Instagram      text   836.0   893.0     443.0  5802.0
...          ...        ...       ...     ...     ...       ...     ...
1995  2023-01-15   Facebook     video  2039.0   523.0     371.0  4780.0
1996  2022-11-13    Twitter     image  1652.0   891.0     410.0  8741.0
1997  2023-09-18    Twitter     video   528.0   207.0     341.0  6588.0
1998  2023-04-23   Facebook     video  1288.0   893.0     328.0  8349.0
1999  2023-08-13    Twitter     image   835.0   532.0     424.0  9236.0

[2000 rows x 7 columns]
```

4. Clean the data by filling missing values (NaN) with 0.

```
df.fillna(0, inplace=True)
```

5. Filter out invalid platforms (keep only Twitter, Instagram, Facebook).

```
valid_platforms=["Twitter","Instagram","Facebook"]
data=df[df["platform"].isin(valid_platforms)]
```

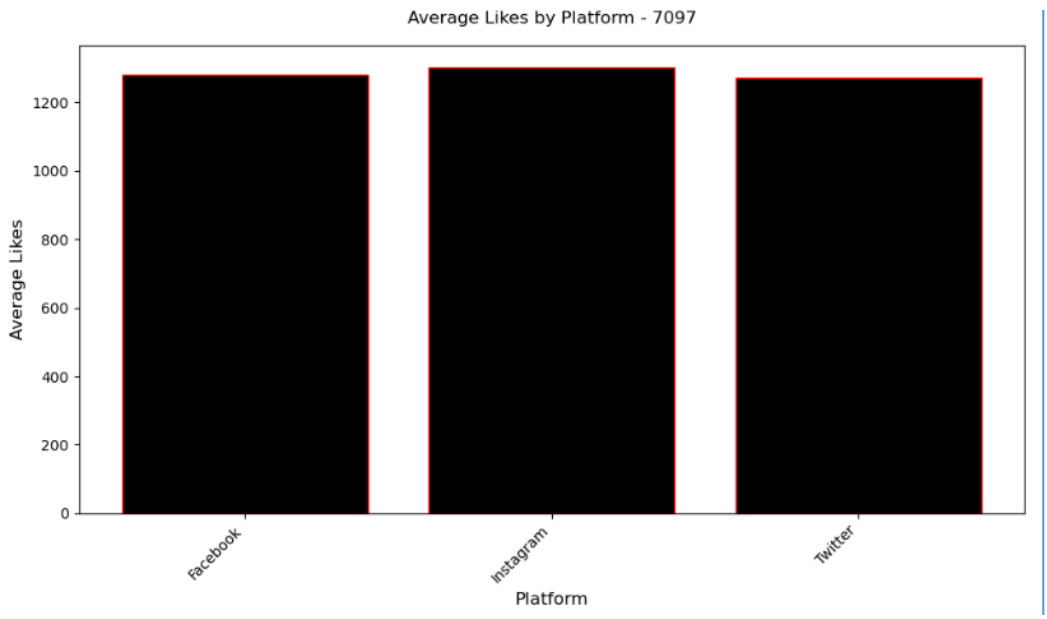
6. Aggregate average metrics (likes, shares, comments, views) by platform.

```
agg_data = df.groupby("platform").agg({
    "likes": "mean",
    "shares": "mean",
    "comments": "mean",
    "views": "mean"
}).reset_index()
print("\nAggregated data:")
print(agg_data)
```

```
Aggregated Data:(7097)
    platform        likes      shares    comments        views
0   Facebook  1279.760697  511.515055  310.667195  6939.965135
1  Instagram  1301.985251  494.438053  298.825959  6875.318584
2    Twitter  1273.132847  481.029197  304.964964  6982.543066
```

7. Create a bar chart for average likes by platform.

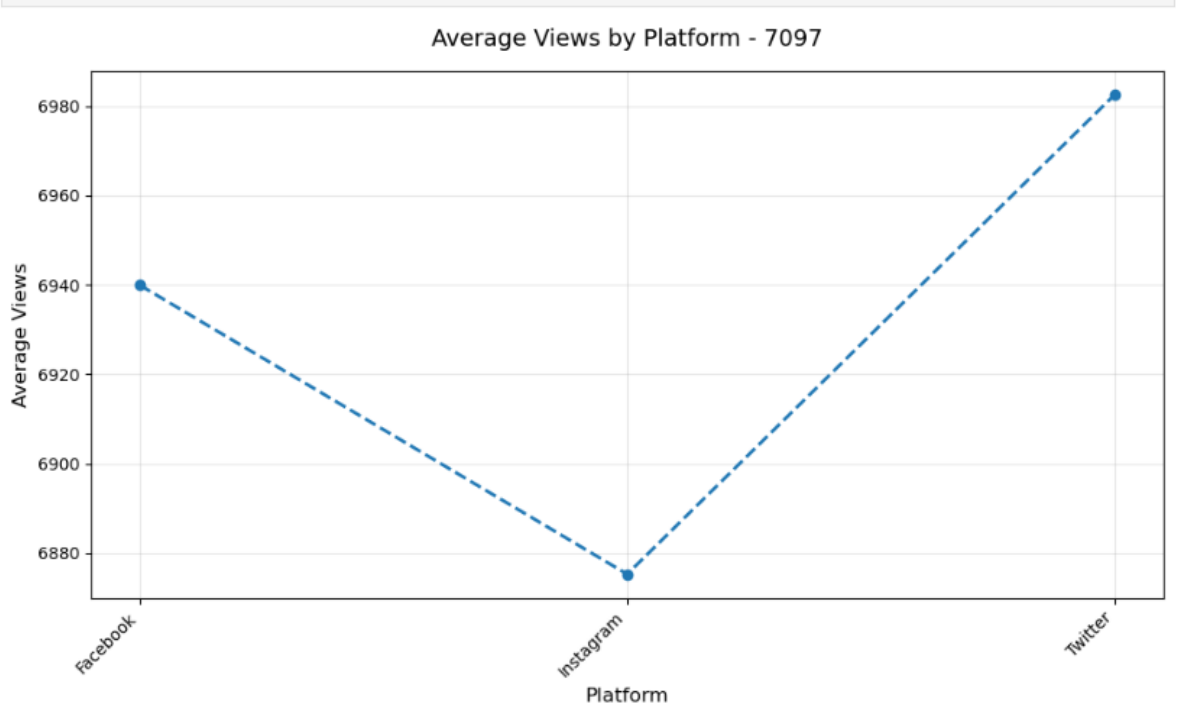
```
plt.figure(figsize=(10, 6))

plt.bar(agg_data['platform'], agg_data['likes'], color='skyblue',edgecolor='black')

plt.title('Average Likes by Platform - 7071',fontsize=12,pad=15)

plt.xlabel('Platform',fontsize=12)

plt.ylabel('Average Likes',fontsize=12)

plt.xticks(rotation=45,ha='right')

plt.tight_layout()

plt.savefig("bar_chart_avg_likes.png", dpi=300)
plt.show()
```



8. Create a line chart for average views by platform.

```
plt.figure(figsize=(10, 6))

plt.plot(agg_data['platform'], agg_data['views'], marker='o', linestyle='--',
    color='green',linewidth=2)

plt.title('Average Views by Platform - 7071',fontsize='14',pad='15')


plt.xlabel('Platform',fontsize=12)

plt.ylabel('Average Views',fontsize=12)
```

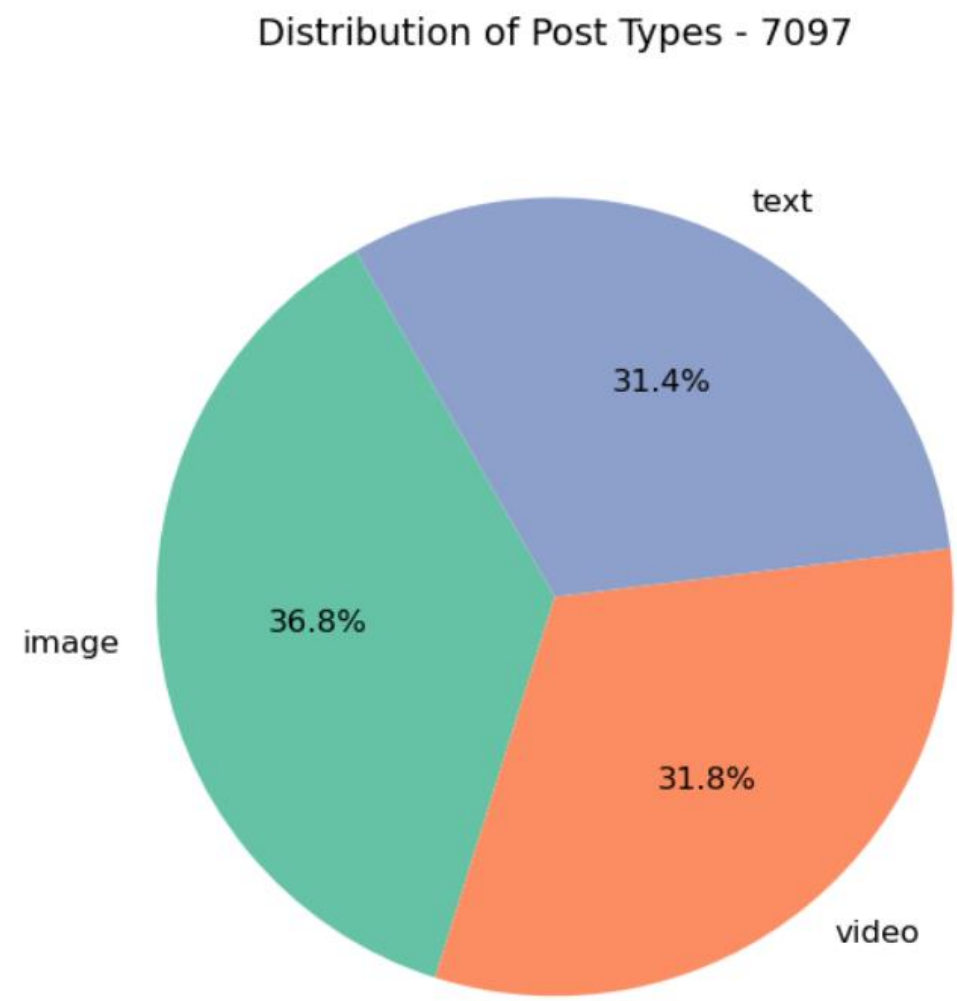
```
plt.xticks(rotation=45,ha='right')

plt.grid(True, alpha=0.3)

plt.tight_layout()

plt.savefig('line_chart_avg_views.png', dpi=300)
plt.show()
```



9. Create a pie chart for the distribution of post types.

```
post_type_counts = data['post_type'].value_counts()

plt.figure(figsize=(6, 6))

plt.pie(post_type_counts,

    labels=post_type_counts.index,

    autopct='%1.1f%%',

    startangle=140,

    colors=plt.cm.Paired.colors,

    textprops={'fontsize': 12})


plt.title('Distribution of Post Types - 7071',fontsize=14,pad=20)

plt.tight_layout()

plt.savefig('pie_chart_post_type.png', dpi=300)
```

```
plt.show()
```



YOUTUBE VIDEO LINK:
https://youtu.be/wYoQAj0KL4U

Result:
The Python script successfully analyzed social media data from a CSV file and generated bar, line, and pie charts that clearly illustrated engagement patterns across platforms and post types.